



Waste Material Segregation for Improving Waste Management

Submitted By: Shivanshu Kumar Singh



Objective

Implement an automated system for waste material classification using Convolutional Neural Networks (CNNs). The aim is to:

- Accurately sort waste into seven categories: **Cardboard**, **Food Waste**, **Glass**, **Metal**, **Other**, **Paper**, and **Plastic**.
- Enhance recycling efficiency and reduce environmental impact.
- Support sustainable waste management by understanding material properties through visual learning.



Dataset Overview

- Image-based dataset structured into **seven folders**, each corresponding to a distinct waste category.
- Each folder contains various images without subcategorization (e.g., `Food_Waste` includes fruit peels, coffee grounds, etc.).
- Uniform image size discovered: **256x256 pixels**, simplifying preprocessing.
- Class imbalance observed (some categories have notably fewer samples).



Data Preparation and Preprocessing



Key Steps

- All images resized to **256x256** for consistency.
- Labels encoded using `LabelEncoder` for integer mapping.
- Applied `image_dataset_from_directory` for efficient loading.
- Dataset split: **6100 training images**, **1525 validation images**.



Visualizations

- Class distribution** visualized using seaborn bar plots.
- Sample images displayed to verify image-label mapping across categories.



CNN Model Architecture



Structure

- Input: (256, 256, 3)
- 3 Convolutional Layers** with increasing filters: 32 → 64 → 128
- Regularization: Batch Normalization + Dropout (0.25 and 0.5)
- Flatten → Dense Layer (128 units) → Output Layer (7 classes with Softmax)

Total Parameters: ~16.8 million

Activation Function: ReLU

Optimizer: Adam

Loss Function: Sparse Categorical Crossentropy



Callbacks Used

- Early Stopping
- Model Checkpointing



Model Training Results



Without Augmentation

- Accuracy improved steadily to **~0.88 (Train)** and **~0.63 (Validation)** by 14th epoch.
- Some overfitting observed after peak performance epochs.
- Validation loss plateaued around **1.25**.



With Augmentation

- Balanced classes using augmentation techniques: random flip, rotation, zoom, contrast, and brightness.
- Model retrained on **12,852 augmented training samples**.
- Validation accuracy climbed to **~0.64**, validating the benefit of balanced data.

Evaluation Metrics

Classification Report Highlights

Class	Precision	Recall	F1-Score
Cardboard	0.63	0.76	0.69
Food Waste	0.75	0.47	0.58
Plastic	0.64	0.63	0.63
Overall Accuracy	0.57		

 **Confusion Matrix** shows misclassifications mostly among visually similar categories (e.g., Glass vs Metal, Paper vs Plastic).

Insights and Conclusions

Findings from Data

- Clear class imbalance, necessitating data augmentation.
- Consistent image sizing aided in streamlined training.
- Visual diversity within each category provides challenges in accurate classification.

Model Results

- CNN with normalization and dropout layers was successful in distinguishing waste types.
- Augmentation significantly improved classification of underrepresented classes.
- Model shows promise for deployment in real-world waste sorting systems.